

2023 JUNIOR CANADIAN CHEMISTRY OLYMPIAD

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

考试期间允许使用化学元素周期表、数据表以及非编程计算器。

Write your answers on the Scantron Sheet provided using a 2B pencil.

请将答案用 2B 铅笔正确填涂在答题卡上。

Students are expected to attempt all questions for a complete examination in 1.5 hours.

考试时长为 1.5 小时。

- 1) The density of carbon tetrachloride at 25 °C is 1.59 g/cm³.

What is the volume of 1 mole of CCl₄ at 25°C ?

温度为 25 °C 时，四氯化碳的密度为 1.59 g/cm³。请问温度为 25 °C 时，1 摩尔 CCl₄ 的体积是多少？

A) 75.1 cm³ B) 9.67 cm³ C) 89.2 cm³ D) 1.03 cm³ E) 96.7 cm³

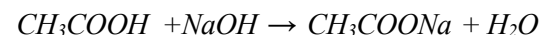
- 2) Eugenol (C₁₀H₁₂O₂) has been shown to have antibacterial, antifungal, antioxidant and antineoplastic activity. Clove oils including eugenol have been claimed to have gentle local anesthetic and antiseptic activities and previously were commonly used in dentistry. Eugenol is the major constituent (80%) in the aromatic oil extract from cloves. How many moles of eugenol are there in 1 kg of clove oil extract?

丁香酚 (C₁₀H₁₂O₂) 已被证明具有抗菌、抗真菌、抗氧化和抗肿瘤活性。含丁香酚的丁香油具有温和的局部麻醉作用和抗菌活性，曾经常用于牙科。丁香酚是从丁香中提取的芳香油的主要成分（80%）。请问 1 千克丁香油提取物中含有多少摩尔的丁香酚？

A) 0.257 B) 4.87 C) 6.09 D) 7.61 E) 13.1

- 3) What is the type of the following reaction?

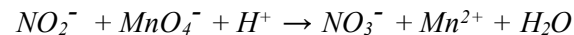
以下属于哪种反应类型？



- A) Synthesis 合成反应
B) Single replacement 单一置换反应
C) Decomposition 分解反应
D) Neutralization 中和反应
E) Combustion 燃烧反应

- 4) After balancing the following reaction, the stoichiometric coefficient of MnO₄⁻ is 2 and the stoichiometric coefficient of NO₂⁻ is 5. What is the stoichiometric coefficient of H⁺?

以下化学反应配平后，MnO₄⁻ 的化学计量系数为 2，NO₂⁻ 的化学计量系数为 5。请问 H⁺ 的化学计量系数是多少？



A) 2 B) 3 C) 4 D) 5 E) 6

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- 5) Which of the following chemical equations is properly balanced?

以下化学方程式配平正确的是哪个?

- A) $\text{C}_6\text{H}_{12}\text{O}_6 + 12\text{O}_2 \rightarrow 6\text{H}_2\text{O} + 6\text{CO}_2$
 B) $\text{C}_6\text{H}_8 + 6\text{O}_2 \rightarrow 5\text{CO}_2 + 4\text{H}_2\text{O}$
 C) $4\text{NH}_2\text{CH}_2\text{COOH} + 9\text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O} + 2\text{N}_2$
 D) $\text{C}_2\text{H}_2 + 2\text{O}_2 \rightarrow 2\text{CO}_2 + \text{H}_2\text{O}$
 E) $\text{C}_{12}\text{H}_{22}\text{O}_{12} + 2\text{H}_2\text{O} \rightarrow 2\text{C}_6\text{H}_{12}\text{O}_6$

- 6) Menthol has local anesthetic and counterirritant qualities, and it is widely used to relieve minor throat irritation. Yearly world production of menthol is about 12,000 tons. Due to its volatility, the qualitative experience of menthol may be modulated by its preparation and combination with other compounds. One such method of preparation is dilution. How many grams of menthol (156.26 g/mol) is needed to prepare a 0.1M solution in ethanol in a volumetric flask of 200 mL?

薄荷醇具有局部麻醉和抗刺激性，被广泛用于缓解轻微的咽喉疼痛，每年薄荷醇的全球产量约为 12,000 吨。由于其挥发性，薄荷醇的定性研究可能会根据它的制备方式和与其他化合物的合成反应来调整。一种制备薄荷醇的方法是通过稀释。在体积为 200 毫升的容量瓶中，需要在乙醇中加入多少克薄荷醇(156.26 g/mol)才能制备出 0.1M 的薄荷醇溶液?

- A) 12.8 B) 7.81 C) 3.13 D) 7.61 E) 13.1

- 7) German chemist Eilhard Mitscherlich discovered in 1844 that solutions of salts of tartaric acid affected polarised light in a surprising way. Residues deposited during wine-making always rotated the plane polarized light to the right while tartaric acid synthesised in the laboratory had no effect on polarized light. Yet the two acids were chemically indistinguishable. Louis Pasteur, born 200 years ago, noticed something Mitscherlich had missed. Pasteur spotted that one salt, sodium ammonium racemate, crystallises in two geometrical forms which are mirror-images of each other. When he separated them (painstakingly picking out individual crystals with

tweezers), and then dissolved them in two separate solutions in water, both solutions rotated the plane of polarised light, in opposite directions! Today, this phenomenon is called optical activity and measuring the specific rotation of optically active compounds is important for organic chemistry. The specific rotation of a compound is calculated as follows:

1844 年，德国化学家艾尔哈德·米切利希发现，酒石酸盐溶液以意想不到的方式影响着偏振光。酿酒过程中残渣沉积物总是使平面偏振光向右旋转，但是实验室合成的酒石酸对偏振光没有影响。然而，这两种酸在化学上是难以区别的。200 年前出生的路易·巴斯德注意到米切利希忽视的一点。巴斯德发现一种盐，即外消旋酒石酸铵钠，结晶为具有两个互为镜像的几何形状。当他将它们分离开来（用镊子费力地挑出单个晶体），然后分别溶解在两个独立的水溶液中时，两个溶液都会使平面偏振光向相反方向旋转！今天，这种现象被称为旋光性，并且有机化学中测量光学活性化合物的比旋度非常重要。化合物的比旋度计算公式如下：

$$[\alpha] = \frac{\alpha}{l[X]}$$

Where $[\alpha]$ is the specific rotation of the compound, α is the rotation of the plane polarized light in degrees, l the path length of the cell in dm and $[X]$ is the concentration of the compound in g/mL. 2.89 g of one of the optically active crystals separated by Pasteur is dissolved in 10 mL of water and placed in a cell of 1 dm, the observed rotation of polarized light is -10° . What is the specific rotation of the crystal?

其中 $[\alpha]$ 是化合物的比旋度， α 是平面偏振光旋转的角度（以度为单位）， l 是旋光管长度（以 dm 为单位）， $[X]$ 是化合物的浓度（以 g/mL 为单位）。将 2.89g 巴斯德分离的一种光学活性晶体溶解在 10 mL 水中，并放入 1 dm 的旋光管中，观测到偏振光旋转 -10° 。请问这种晶体的比旋度是多少？

- A) -34.6° B) $+28.9^\circ$ C) -28.9° D) -10° E) $+34.6^\circ$

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- 8) Three acids that can be found in a household are acetic acid (AA), citric acid (CA), and bisulfate (BSA). The pK_a values of the three acids are 4.76, 2.79, and 1.92, respectively. Which option correctly ranks these acids in the order of **decreasing** acid strength?

在家里能找到的三种酸分别是醋酸 (AA)、柠檬酸 (CA) 和硫酸氢盐 (BSA)。这三种酸的 pK_a 值分别为 4.76、2.79 和 1.92。以下哪项正确按照酸的强度**从高到低**排列？

- A) AA > CA > BSA
- B) CA > BSA > AA
- C) AA > BSA > CA
- D) BSA > CA > AA
- E) BSA > AA > CA

- 9) Ultraviolet/Visible (UV/Vis) spectroscopy is routinely used in analytical chemistry for the quantitative determination of diverse analytes. In a typical UV/Vis experiment, attenuation is measured for light traveling through a sample solution. The light attenuation can be used to calculate the absorbance of the solution. The Beer–Lambert law relates the absorbance to the molar absorptivity of the molecule, the path length of the light and the concentration of the solution:

紫外线/可见光 (UV/Vis) 光谱常出现在分析化学中，用于对不同分析物进行定量测定。典型的紫外线/可见光实验中，穿过样品溶液的光出现衰减。光衰减可用于计算溶液的吸光度。比尔-朗伯定律 (Beer–Lambert law) 将吸光度与分子的摩尔吸光系数、光程和溶液的浓度联系在一起：

$$A = \epsilon lc$$

Where A is absorbance, ϵ is the molar absorptivity (a property of the molecule), and l is the path length of the light through the sample and c the concentration of the solution.

What is the concentration for a Blue #1 dye solution with

an absorbance of 0.49 and a molar absorptivity coefficient of $130,000 \text{ M}^{-1}\text{cm}^{-1}$, measured in a cuvette with a path length of 0.25 cm at 630 nm?

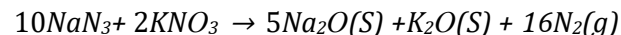
其中 A 表示吸光度， ϵ 表示摩尔吸光系数（分子的一种性质）， l 表示光通过样品的光程， c 表示溶液的浓度。

在距离 630 nm 的位置用路径长度为 0.25 cm 的试管进行测量，吸光度为 0.49 且摩尔吸光度系数为 $130,000 \text{ M}^{-1}\text{cm}^{-1}$ 的蓝色 1 号染料溶液的浓度是多少？

- A) 15 μM B) 3.8 nM C) 15 mM D) 0.94 μM E) 3.8 μM

- 10) Airbags contain a mixture of sodium azide, potassium nitrate, and silicon dioxide. A sensor detects a head on collision that cause the sodium azide to be ignited and to decompose forming sodium oxide and nitrogen gas as follows:

安全气囊中混合有叠氮化钠、硝酸钾和二氧化硅。传感器会检测到正面碰撞，导致叠氮化钠燃烧并分解，生成氧化钠和氮气，其化学反应如下：



The average volume of an airbag is 140 L. Assuming that the airbag is filled at 1 atm at a temperature of 298 K, how many moles of NaN_3 are needed on average? (assume ideal gas behavior)

气囊的平均体积为 140 L，假设气囊在一个标准大气压 (1 atm) 且温度为 298 K 的条件下被充满，请问平均需要多少 mol 的 NaN_3 ？（假设符合理想气体的性质）

- A) 5.73 B) 3.58 C) 3.91 D) 17.7 E) 25.9

- 11) Which of the following atoms has the largest first ionization energy?

下面哪个原子的第一电离能最大？

- A) Si B) F C) K D) Be E) Cl

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12) Which of the following has the greatest bond order?

以下哪项键级最大?

- A) O_2^{2+} B) O_2^+ C) O_2 D) O_2^- E) O_2^{2-}

13) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value. A good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$, where Z is the atomic number of the atom and S a screening constant. In the second row of the periodic table, the screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons, excluding the electron for which the effective nuclear charge is to be determined. Use Slater's rule to estimate the Z_{eff} for a valence electron in neon (Ne).

有效核电荷 (Z_{eff}) 是理解周期性的有用工具。 Z_{eff} 是电子在
多电子原子中的净正电荷。使用术语“有效”是因为原子核的
正电荷不能完全到达给定的电子，这是由于其他电子的屏蔽
效应。 Z_{eff} 是一个理论的计算值， Z_{eff} 的理想估计值可以使
用斯莱特法则获取。斯莱特法则规定 $Z_{\text{eff}} = Z - S$ ，其中 Z 是
原子的原子序数， S 是屏蔽常数。在周期表的第二行，内层
电子的屏蔽常数为 0.85，价电子为 0.35。屏蔽常数 (S) 通过
计算其他电子 (不包括需要确定有效电荷数的电子) 的所有
屏蔽常数之和得到。请使用斯莱特法则估算氖 (Ne) 中价电
子的有效电荷数。

- A) 3.35 B) 5.85 C) 2.35 D) 6.85 E) 4.15

14) While preparing a solution of KNO_3 in water, a chemistry student noticed that the volumetric flask felt cold. Intrigued, the student decided to perform a rudimentary experiment to measure the enthalpy of solution (ΔH) of KNO_3 . For his experiment, the student added 10.1g KNO_3 to 180g of water and used a thermometer to record the temperature. A temperature change of -4.6 degrees was recorded. Assuming that the heat capacity of water is unchanged at $75 \text{ JK}^{-1}\text{mol}^{-1}$, what is the enthalpy of solution for KNO_3 ?

在水中制备 KNO_3 溶液时，一名化学学生注意到容量瓶摸
起来较冷。学生对此感到好奇，决定进行一项基本实验来
测量 KNO_3 的溶解热 (ΔH)。为了完成这项实验，学生
将 10.1g 的 KNO_3 加入到 180 g 水中，并使用温度计记录
温度。记录到温度变化为 -4.6 度。假设水的热容量在 75
 $\text{JK}^{-1}\text{mol}^{-1}$ 时保持不变，请问 KNO_3 的溶解热是多少?

- A) -34.5 kJ/mol
B) 62.1 kJ/mol
C) 34.5 kJ/mol
D) -6.15 kJ/mol
E) -62.1 kJ/mol

15) An element X forms compounds with chlorine as dichloride ($XC\text{l}_2$), which is a white crystalline solid, and tetrachloride ($XC\text{l}_4$), a colorless liquid. Treatment of 10.00 g $XC\text{l}_2$ with excess chlorine forms 13.74 g $XC\text{l}_4$. Identify X.

一种元素 X 与氯可以形成二氯化物 ($XC\text{l}_2$) 和四氯化物
($XC\text{l}_4$)，二氯化物是一种白色结晶固体，四氯化物是一种
无色液体。将 10.00 克 $XC\text{l}_2$ 与过量的氯气反应生成 13.74
克 $XC\text{l}_4$ 。请确定元素 X。

- A) Pt B) Ti C) S D) Sn E) Rh

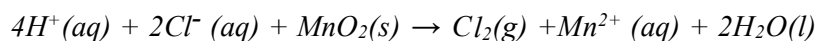
- 16) Following the VSEPR theory, which of the following molecules has a molecular geometry that is trigonal pyramidal?

根据价层电子对互斥理论（VSEPR），以下哪个分子的几何构型是三角锥？

- A) $[\text{BrO}_4]^-$
- B) $[\text{BrO}_3]^-$
- C) $[\text{BrO}_2]^-$
- D) BrO_2
- E) Br_2O

- 17) Which species is the reducing agent in the following reaction?

以下化学反应中，哪种物质是还原剂？

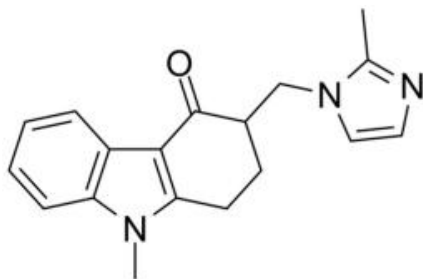


- A) H^+
- B) Mn^{2+}
- C) MnO_2
- D) Cl_2
- E) Cl^-

- 18) The drug shown below is a medication used to prevent nausea and vomiting caused by cancer chemotherapy, radiation therapy, or surgery. Which of the following molecular formula is the correct one for the antiemetic agent drug shown below?

下面所示的药物用于防止由癌症化疗、放射疗法或手术引起的恶心和呕吐。以下哪项是该止吐药物的正确分子式？

- A) $\text{C}_{18}\text{H}_{18}\text{N}_3\text{O}$
- B) $\text{C}_{17}\text{H}_{16}\text{N}_3\text{O}$
- C) $\text{C}_{16}\text{H}_{13}\text{N}_3\text{O}$
- D) $\text{C}_{19}\text{H}_{18}\text{N}_3\text{O}$
- E) $\text{C}_{18}\text{H}_{19}\text{N}_3\text{O}$



- 19) Which of the following is the correct order of increasing melting point?

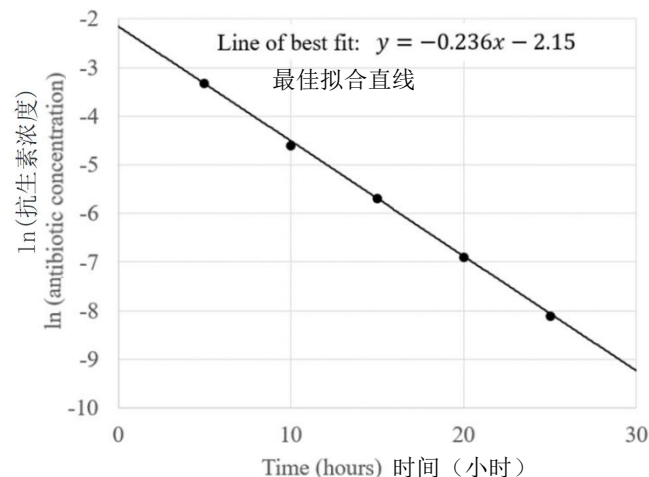
以下哪项按照熔点从低到高的顺序正确排列？

- A) $\text{NaCl}(\text{s})$, $\text{C}_{16}\text{H}_{34}(\text{s})$, $\text{MgO}(\text{s})$, $\text{C}(\text{s})$, $\text{Al}(\text{s})$
- B) $\text{C}_{16}\text{H}_{34}(\text{s})$, $\text{Al}(\text{s})$, $\text{NaCl}(\text{s})$, $\text{MgO}(\text{s})$, $\text{C}(\text{s})$
- C) $\text{C}_{16}\text{H}_{34}(\text{s})$, $\text{C}(\text{s})$, $\text{MgO}(\text{s})$, $\text{NaCl}(\text{s})$, $\text{Al}(\text{s})$
- D) $\text{C}(\text{s})$, $\text{C}_{16}\text{H}_{34}(\text{s})$, $\text{NaCl}(\text{s})$, $\text{Al}(\text{s})$, $\text{MgO}(\text{s})$
- E) $\text{C}_{16}\text{H}_{34}(\text{s})$, $\text{MgO}(\text{s})$, $\text{C}(\text{s})$, $\text{Al}(\text{s})$, $\text{NaCl}(\text{s})$

- 20) The following plot records the process of the elimination of an antibiotic by the human body. What is the half life of the antibiotic?

下图记录了一种抗生素被人体排出的过程。这种抗生素的半衰期是多少？

- A) 4.24 h
- B) 0.24 h
- C) 2.94 h
- D) 0.34 h
- E) 2.15 h



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21) What is the pH of a 0.350 M solution of sodium citrate at 25°C (pK_b of citrate = 7.60)?

温度为 25°C 时, 0.350 M 柠檬酸钠溶液的 pH 值是多少?

(柠檬酸盐的 pK_b 值为 7.60)

A) 4.03 B) 9.97 C) 8.45 D) 5.19 E) 3.28

22) The atomic radius of Zirconium (Zr) is 155 pm. The atomic radius of Hafnium (Hf) is also 155 pm. Which of the following is the best explanation of this phenomenon? (*Hint: look at the periodic table and write down the electron configuration. Slater's rule does not apply to atoms lower than the fourth row*)

锆 (Zr) 和铪 (Hf) 的原子半径都为 155 pm, 以下哪项是对这一现象的最佳解释? (提示: 可以查看元素周期表并写出电子构型。斯莱特法则不适用于排列在第四行以下的原子)

A) The ionization energy of Hf is lower than the ionization energy of Zr. 铪的电离能低于锆的电离能。

B) Hafnium has more protons, therefore the attraction to the valence electrons from the nucleus is lower.

铪的质子数更多, 因此原子核对价电子的吸引力更小。

C) Hafnium has a greater effective nuclear charge and therefore the valence electrons experience stronger attraction from the nucleus.

铪具有更大的有效核电荷, 因此价电子受到原子核的吸引力更强。

D) Zirconium has a greater effective nuclear charge and therefore the valence electrons experience stronger attraction from the nucleus.

锆具有更大的有效核电荷, 因此价电子受到原子核的吸引力更强。

E) The highest sub-energy level in an atom of Zirconium is 5s while in an atom of Hafnium is 6s.

锆原子中最高的亚能级是 5s, 而铪原子中最高的亚能级是 6s。

23) A liquid organic compound X has a molar mass of less than 150 g/mol. A sample of 11.8 g of this compound completely combusted in 20.0 L of oxygen to produce carbon dioxide and water vapor. After the reaction, the gas phase of the reaction mixture was passed through a concentrated H₂SO₄(aq) solution (H₂SO₄(aq) is a good dehydrating agent). The remaining gas had a volume of 17.76 L. This gas was then passed through solid CaO, which reduced the volume of the gas by 11.20 L. Identify X. (assume all reactions were complete and all gas volumes are based on a pressure of 1 atm and a temperature of 273.15K).

一种液态有机化合物 X 的摩尔质量小于 150 g/mol。11.8 g 该化合物样品在 20.0 L 的氧气中完全燃烧, 产生二氧化碳和水蒸气。发生化学反应后, 反应混合物的气相通过浓硫酸溶液 (浓硫酸是一种很好的脱水剂), 剩余气体的体积为 17.76 L。然后该气体通过固体 CaO, 气体体积减少 11.20 L。请确定化合物 X (假设所有反应都为完全反应, 所有气体的体积都基于 1 atm 和温度为 273.15 K 的条件)。

A) C₆H₁₂O₃ B) C₆H₁₀O₄ C) C₅H₁₀O₃

D) C₆H₁₄O₂ E) C₇H₁₂O₂

- 24) A gas contains hydrogen and other molecules under an atmospheric pressure of 745 mmHg and a temperature of 120.2°C. what can be said about the kinetic energy?
在大气压为 745 mmHg 、温度为 120.2°C 的条件下，一种气体含氢分子和其他分子。关于分子的动能可以推断出什么？

- A) All molecules in the gas will have a lower kinetic energy than hydrogen molecules.
气体中所有分子的动能都比氢分子更小。
- B) All molecules in the gas will have a kinetic energy of zero.
气体中所有分子的动能都为零。
- C) All molecules in the gas will have a different kinetic energy due to their different molar mass.
气体中所有分子的动能不同是因为它们的摩尔质量不同。
- D) All molecules in the gas will have exactly the same kinetic energy.
气体中所有分子的动能恰好相同。
- E) All molecules in the gas will have the same average kinetic energy.
气体中所有分子的平均动能相同。

- 25) Esterification is an important organic reaction. Using acid as a catalyst, which of the following will result in an esterification reaction to produce ethyl acetate.
酯化反应是一种重要的有机反应。下列哪项使用酸作为催化剂，可以发生酯化反应生成乙酸乙酯？

- A) $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{COCH}_3$
- B) $\text{CH}_3\text{CHO} + \text{CH}_3\text{COOH}$
- C) $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{OCH}_3$
- D) $\text{CH}_3\text{COOH} + \text{CH}_3\text{OCH}_3$
- E) $\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH}$

JUNIOR CANADIAN CHEMISTRY OLYMPIAD																							
Periodic table and data sheet																							
1 H 1.008 Hydrogen																	2 He 4.003 Helium						
3 Li 6.941 Lithium	4 Be 9.012 Beryllium																	5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium																	13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton						
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.89 yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc (98) Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon						
55 Cs 132.9 Cesium	56 Ba 137.3 Barium	57 La 138.9 Lanthanum	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po (209) Polonium	85 At (210) Astatine	86 Rn (222) Radon						
87 Fr (223) Francium	88 Ra (226) Radium	89 Ac (227) Actinium	104 Rf (261) Rutherfordium	105 Db (262) Dubnium	106 Sg (266) Seaborgium	107 Bh (264) Bohrium	108 Hs (277) Hassium	109 Mt (268) Meitnerium	110 Ds (269) Darmstadtium	111 Rg (272) Roentgenium	112 Cn (285) Copernicium	113 Nh (284) Nihonium	114 Fl (289) Flerovium	115 Mc (288) Moscovium	116 Lv (292) Livermorium	117 Ts (294) Tennessine	118 Og (294) Oganesson						

58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm (145) Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu (244) Plutonium	95 Am (243) Americium	96 Cm (247) Curium	97 Bk (247) Berkelium	98 Cf (251) Californium	99 Es (252) Einsteinium	100 Fm (257) Fermium	101 Md (258) Mendelevium	102 No (259) Nobelium	103 Lr (262) Lawrencium

原子质量单位	Atomic mass unit	1.66054×10 ⁻²⁷ kg
阿伏伽德罗常量	Avogadro's number	6.022×10 ²³
电子电荷量	Charge of an electron	1.60218×10 ⁻¹⁹ C
法拉第常数	Faraday's constant	96485 C mol ⁻¹
理想气体常数	Gas constant	8.3145 J K ⁻¹ mol ⁻¹
电子质量	Mass of an electron	9.1094×10 ⁻³¹ kg
光速	Speed of light	2.99792458×10 ⁸ ms ⁻¹
普朗克常数	Plank Constant	6.62637×10 ⁻³⁴ Js

1 atm = 101325 Pa
1 bar = 100000 Pa
0 °C = 273.15 K